in different survival rates and reproductive abilities. Those organisms that have higher breeding success pass on their genes to more of the next generation, so individuals with these characteristics become more common.

“Natural selection is daily and hourly scrutinizing, throughout the world, the slightest variations.”

Charles Darwin

To the Galápagos

The young Charles Darwin first began to consider evolution during his pioneering scientific expedition around the world aboard HMS *Beagle* from 1831 to 1836. As a young man, Darwin accepted the orthodox interpretation of the Bible, that Earth was only a few thousand years old. However, while he was on board the *Beagle*, Darwin read Scottish geologist Charles Lyell’s recently published *Principles of Geology*, in which Lyell demonstrated that rocks bore traces of tiny, gradual, and cumulative change over vast time periods—millions, rather than thousands of years. As Darwin looked at landscapes around the world that had been affected by processes of erosion, deposition, and volcanism, he began to speculate about animal species changing over very long time periods, and the reasons for such changes. By examining fossils and observing living animals, Darwin identified patterns; he noticed, for example, that extinct species had often been replaced by similar, but distinct, modern ones.

Darwin’s field work on the islands of the Galápagos archipelago off South America in the fall of 1835 provided especially strong evidence for his later theory of evolution by natural selection. Here, he observed that the shape of the carapaces (shells) of giant tortoises varied slightly from island to island. Darwin was also intrigued to find that there were four broadly similar, yet clearly distinct, varieties of mockingbirds, but that no single island had more than one species of the bird. He saw small birds, too, that